

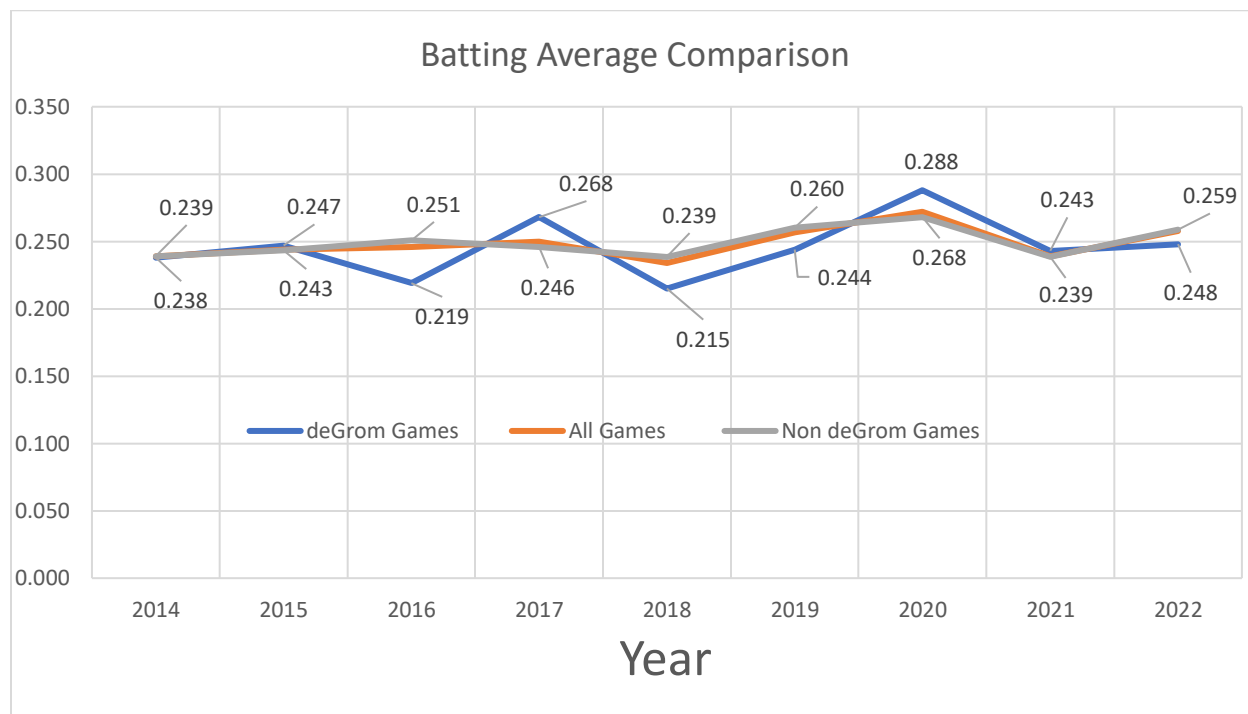
Is There a deGrom “Curse”?

By: Jack Burns*

Back in 2021, I was watching Gary, Ron, and Keith of SNY lament the lack of Mets offense during another superb outing by Mets ace Jacob deGrom. While I don’t remember the game in question, what did stick with me was the counterintuitive observation that the Mets seemed to perform worse with deGrom on the mound. As their ace, deGrom games should be walks in the literal park for the Mets, but the Mets record when deGrom pitches is only 113 – 94. In other words, the Mets only win about 55% of the time when deGrom pitches, not much better than a coin flip. So maybe the guys are right, maybe there is something here.

With a statistics degree going to waste and time on my hands, I decided to look into this. Would the data back up this observation? Or is it just a Mets fan’s conditioning to expect the worst?

To investigate this, I gathered data from the [baseball-reference.com](https://www.baseball-reference.com)¹ for all the Mets games since deGrom’s debut back in 2014². I looked at two metrics: Batting Average, and Run to Hit Ratio (i.e., how many runs did the Mets manufacture from their hits). First, I compared the Mets overall batting average to their batting average when deGrom was pitching:



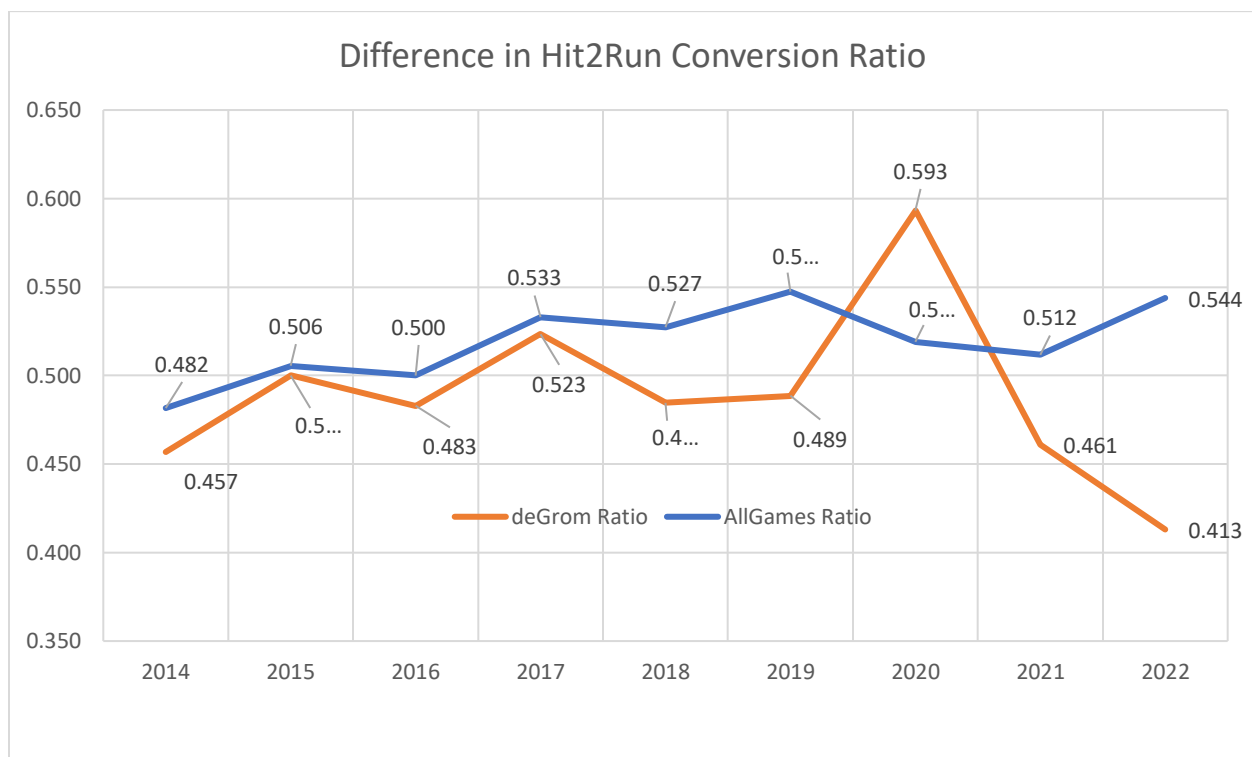
As the chart above shows, the Mets batting average varies greatly for deGrom season by season. The high point was the shortened 2020 season where they hit .288 behind deGrom, and the low point was the 2018 season where they hit only .215 (there was a stretch of 16 games (5/13/2018 – 8/3/2018) where the Mets only hit above .200 four times with deGrom on the mound). Just to

¹ <https://www.baseball-reference.com/teams/NYM/batteam.shtml>

² <https://www.baseball-reference.com/players/gl.fcgi?id=degroja01&t=p&year=2014>

confirm that it wasn't that the Mets were not hitting when deGrom pitched, I flexed my statistics muscles and used the Student's t-test. This test is used to determine statistical differences in small samples, which works well here because we only have a few seasons of data to compare (Fun fact: the Student in the Student's t-test does not refer to actual students but a pseudonym used by William Sealy Gosset, a worker at the Guinness brewery who published work into the issues of working with small sample sizes in the early 20th century³). The Student's t-test showed that there was no significant difference between the means of the batting averages in the games deGrom pitched and the games he did not. This means that the Mets do hit at the same rate behind deGrom as they do with other pitchers, so I needed to keep looking.

Since batting average wasn't the issue, I delved into the Run to Hit Ratio. I wanted to see if there was a significant difference in how many runs were scored off the number of hits the Mets had. Overall, the Mets have a 0.519 Run to Hit Ratio for the years deGrom played meaning that on average, they needed less than two hits to get one run. However, when looking into the games deGrom pitched that ratio decreased 0.489, meaning that the Mets, on average, would need more than two hits to get one run. While deGrom's Run to Hit Ratio has always been lower than the rest of the Mets pitching, this difference has gotten more pronounced over time with increasing to a 0.042 difference on average for the last five years, shown below:



If we remove the 2020 season as an outlier, then the difference is over 0.07 on average, and with this current season at a massive .131 difference. This means that, on average, in games that deGrom pitched in 2022, the Mets needed to get 2.46 hits to get one run, compared to only 1.89 hits to get one run for the 2022 season overall.

³ https://en.wikipedia.org/wiki/Student%27s_t-test

Now, the important question, is this statistically significant? Well, when looking at deGrom's overall career, the t – test found no significant difference. However, when looking at only complete seasons that deGrom played (2015 – 2019), the results showed that the difference was in fact statistically significant at the 5% level! This means we can say with 95% confidence that the two means of the samples are different (this 95% being the typical cut off for statistical analyses). In plain English, the Mets did have a decreased ability to convert their hits to runs when deGrom pitches compared to other Mets pitchers during the years he pitched complete seasons. Additionally, once I removed the short 2020 season as an outlier, and compared all the seasons deGrom pitched, the difference between the Non deGrom games and the deGrom games Run to Hit Ratio was significant at the 1% level, showing even greater confidence that the means of the two are not the same.

Digging a little deeper, I wanted to see if the reason why Mets have a low Run to Hit ratio with deGrom is because the opposing team puts an ace pitcher on the mound against them to balance out the game. This ace pitcher could get out of jams and strand Met runners on base easily, thus decreasing the ratio. To find out, I cross-referenced deGrom's games against the fangraphs.com⁴ page on top pitchers per season, counting anyone in the top 30 for that year as an ace. After grouping the games by season and if the Mets were up against an ace, I ran another t-test and found (drum roll please), nothing, absolutely nothing, there was no statistical difference in batting average or in Run to Hit ratio when deGrom versed another ace.

Lastly, I wanted to see if this “curse” extended to another Mets ace, Max Scherzer. As Scherzer has only played for the Mets for one season, 2022, a t-test would not apply here, so we can only examine the numbers. Interestingly, whatever team mentality the Mets had with deGrom seems to have also passed to Scherzer, as both the Mets batting average (.237) and Run to Hit ratio (0.455) with Scherzer are lower than the overall (.259 and 0.538, respectively).

But what does this mean? Well, it shows that Gary, Ron, and Keith were correct, as they always are. The Mets do in fact have an issue manufacturing runs when deGrom or Scherzer is on the mound. The players themselves might not even be aware of it, subconsciously relaxing knowing their aces will dominate. With the playoffs coming up and the Mets in it this year, I felt that it's something interesting to research. As a lifelong Mets fan, I can't wait to watch the Mets play this post season and who knows maybe I'm completely wrong and the Mets crush all that stand in their way with deGrom on the mound. Let's Go Mets!

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⁴<https://www.fangraphs.com/leaders.aspx?pos=all&stats=pit&lg=all&qual=y&type=8&season=2014&month=0&season1=2014&ind=0>